

Histogram Equalization Using OpenCV (Grayscale & Color Images)

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```
In [2]: import cv2  
        from matplotlib import pyplot as plt
```

```
In [3]: # Load the color image  
image = cv2.imread('parrot.jpg')
```

```
In [4]: # Convert the image to grayscale  
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
In [5]: plt.imshow(gray_image, cmap='gray')  
plt.title('Original Grayscale Image')  
plt.axis('off')
```

```
Out[5]: (-0.5, 1279.5, 852.5, -0.5)
```

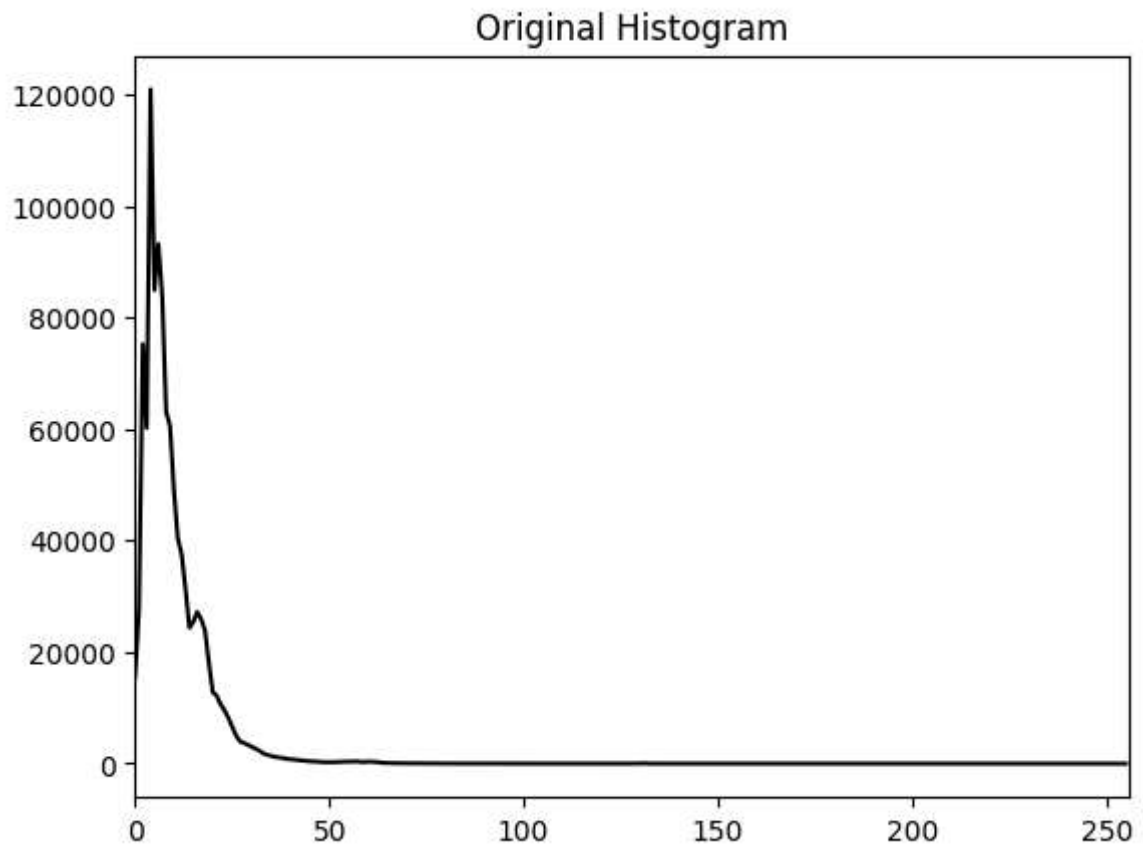
Original Grayscale Image



```
In [6]: hist_original = cv2.calcHist([gray_image], [0], None, [256], [0, 256])
```

```
In [7]: plt.plot(hist_original, color='black')  
plt.title('Original Histogram')  
plt.xlim([0, 256])
```

Out[7]: (0.0, 256.0)



```
In [8]: # Apply histogram equalization  
equalized_image = cv2.equalizeHist(gray_image)
```

```
In [9]: plt.imshow(equalized_image, cmap='gray')  
plt.title('Equalized Image')  
plt.axis('off')
```

```
Out[9]: (-0.5, 1279.5, 852.5, -0.5)
```

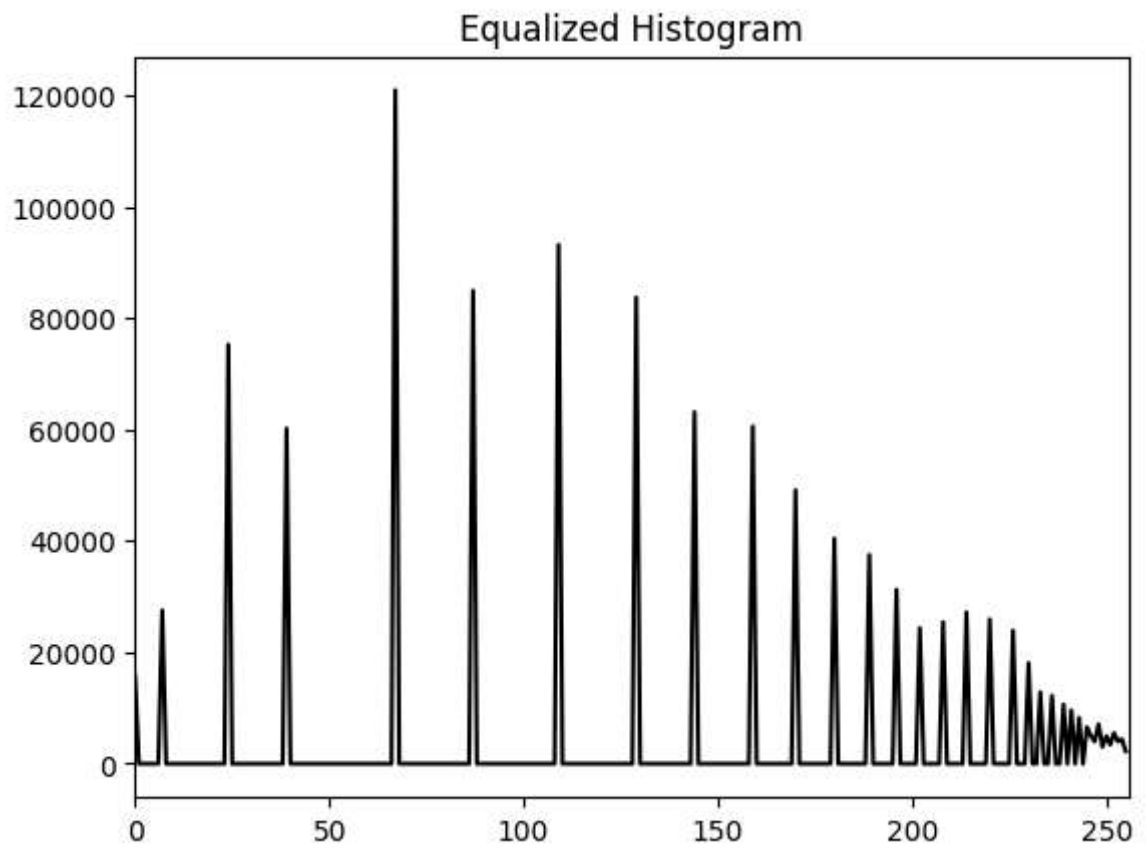
Equalized Image



```
In [10]: hist_equalized = cv2.calcHist([equalized_image], [0], None, [256], [0, 256])
```

```
In [11]: plt.plot(hist_equalized, color='black')  
plt.title('Equalized Histogram')  
plt.xlim([0, 256])
```

Out[11]: (0.0, 256.0)



In []: